SYSTEM REQUIREMENTS SPECIFICATION (SHEET 1)

1. FUNCTIONAL REQUIREMENTS MATRIX

ID	REQUIREMENT DESCRIPTION	ACCEPTANCE CRITERIA	ENGINEERING IMPLEMENTATION
FR-01	Stream continuous H.264 compressed video over local Wi-Fi without frame stutter.	30.0 fps sustained, zero dropped frames over 30 min.	Rockchip RV1103 Hardware VPU Encoder.
FR-02	Minimum native video capture resolution.	1920 x 1080 (1080p Full HD).	SmartSens SC3336 3MP Sensor + ISP 3.2.
FR-03	Zero external SD card dependency for system boot and runtime execution.	Successful system cold boot with SD slot empty.	128MB SPI NAND Flash (Luckfox Mini B).
FR-04	Wireless local point-to-point network connectivity (SoftAP).	Phone associates directly without external router.	RTL8188EUS USB Wi-Fi (HostAPD 802.11n).
FR-05	Battery-buffered autonomous operation during engine cut-off.	Greater than 60 minutes continuous streaming.	3.7V 1000mAh LiPo Pack with PCM Protection.
FR-06	Simultaneous vehicle power pass-through and battery recharge.	Battery charges while system streams video.	IP5306 Synchronous PMIC with Load-Sharing.
FR-07	Automotive electrical transient and overvoltage protection.	Clamps inductive spikes up to 40V safely.	SMBJ5.0A TVS Diode + 470uF Bulk Reservoir.
FR-08	Power management auto-sleep inhibition circuit.	PMIC maintains continuous 5V output.	100 Ohm Resistor (R1) static ~50mA pull.
FR-09	Wide-angle forward optical field of view.	Diagonal FOV greater than 95.0 degrees.	98.3 degree wide-angle lens assembly.
FR-10	Visual hardware status indication.	Optical indication of streaming state.	GPIO driven Status LED via 470 Ohm resistor.

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		ENGINEERING DISCIPLINE SYSTEMS ENGINEERING	STATUS RELEASED	

SYSTEM REQUIREMENTS SPECIFICATION (SHEET 2)

2. NON-FUNCTIONAL PERFORMANCE & ENVIRONMENTAL SPECIFICATIONS

METRIC ID	PERFORMANCE METRIC	SPECIFICATION LIMIT	VERIFICATION TECHNIQUE
NFR-01	Total Bill of Materials Cost	Less than INR 3,800.00 (Actual: INR 3,264.00)	Procurement reconciliation audit.
NFR-02	Cold Boot Duration to Video Stream	Less than 6.0 Seconds from cold power applied	Serial timestamp measurement.
NFR-03	End-to-End Glass-to-Glass Latency	Less than 250 Milliseconds	High-speed stopwatch comparison.
NFR-04	Ambient Operating Temperature Envelope	0 Deg C to 45 Deg C (Non-condensing)	Thermal chamber soak.
NFR-05	Maximum Physical Enclosure Envelope	70 mm (L) x 45 mm (W) x 30 mm (H)	Digital caliper mechanical check.

3. SCOPE BASELINE & EXCLUSIONS

- **No On-Device SD Recording:** SD card elimination avoids write degradation and reduces BOM cost.
- **Client-Side Storage:** Video streams are recorded directly inside client mobile app (e.g. VLC) on demand.

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-REQ-001	DATE 2026-10-08	SHEET 2 OF 2	
DOCUMENT TITLE SYSTEM REQUIREMENTS (SRS)	REV 1.4	DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824		
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